

## M003 Extension — Build the Spider (Intermediate)

**Topic:** 3D Spider (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 6) ·

**Time:** about 40 minutes

Build a **red spider** in 3D — a fat body, eight legs spread out, two eyes on the front. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

● Stand on your **home spot** (red block) every run. Move your feet, move your spider.

🎨 Red parts = **red concrete**. Eyes = **black wool**.

### 1 · Predict 🧠

Read each set. Write a sentence for each.

```
place red block at (1, 1, 2)
place red block at (1, 1, 3)
place red block at (1, 1, 4)
```

**Only z changes. What happens to this leg?**

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```
place red block at (1, 1, 2)
place red block at (5, 1, 2)
```

**Only x changes. Where do these two legs sit?**

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```
place red block at (3, 2, 3)
place red block at (3, 3, 3)
```

**Only y changes. What happens here?**

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## 2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

**Bug A** — This eye should sit **up high on the front**, at (2, 3, 3). It has only two numbers, so it drops flat on the floor.

```
place black block at (2, 3)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work? (1–2 sentences)**

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**Bug B** — This right-side leg should be **across on the right**: (5, 1, 3). The last two numbers are swapped.

```
place red block at (5, 3, 1)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work? (1–2 sentences)**

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### 3 · Show Your Work 📷 🎥

Look at the 🕷️ spider in the world. Walk around it. Then build **one full spider** on your home spot — body, four left legs, four right legs, two eyes.

**Red body:** fill from (2, 2, 3) to (4, 3, 4).

**Left legs** (share x = 1): red block at (1, 1, 2), (1, 1, 3), (1, 1, 4), (1, 1, 5).

**Right legs** (share x = 5): red block at (5, 1, 2), (5, 1, 3), (5, 1, 4), (5, 1, 5).

**Two black eyes on the front:** (2, 3, 3) and (4, 3, 3).

💡 Tip: legs reach **across** (x) to the sides and spread **deeper** (z) front to back. They stay low (small y).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera at the spider. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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**Submit** 

Send this worksheet + your **one** video to teacher on KakaoTalk.